Remote Sensing of Environment 115 (2011) 1301–1316

Contents lists available at ScienceDirect

Remote Sensing of Environment

journal homepage: www.elsevier.com/locate/rse

Object-based crop identification using multiple vegetation indices, textural features and crop phenology

José M. Peña-Barragán *, Moffatt K. Ngugi, Richard E. Plant, Johan Six

Department of Plant Sciences, University of California, Davis, CA 95616, United States

ARTICLE INFO

Article history:
Received 28 August 2010
Received in revised form 14 January 2011
Accepted 16 January 2011
Available online 25 February 2011

Keywords:
ASTER
OBIA
Spectral and textural features
Decision tree
Field-scale
Agriculture

ABSTRACT

Crop identification on specific parcels and the assessment of soil management practices are important for agro-ecological studies, greenhouse gas modeling, and agrarian policy development. Traditional pixel-based analysis of remotely sensed data results in inaccurate identification of some crops due to pixel heterogeneity, mixed pixels, spectral similarity, and crop pattern variability. These problems can be overcome using object-based image analysis (OBIA) techniques, which incorporate new spectral, textural and hierarchical features after segmentation of imagery. We combined OBIA and decision tree (DT) algorithms to develop a methodology, named Object-based Crop Identification and Mapping (OCIM), for a multi-seasonal assessment of a large number of crop types and field status.

In our approach, we explored several vegetation indices (VIs) and textural features derived from visible, near-infrared and short-wave infrared (SWIR) bands of ASTER satellite scenes collected during three distinct growing-season periods (mid-spring, early-summer and late-summer). OCIM was developed for 13 major crops cultivated in the agricultural area of Yolo County in California, USA. The model design was built in four different scenarios (combinations of three or two periods) by using two independent training and validation datasets and the best DTs resulted in an error rate of 9% for the three-period model and between 12 and 16% for the two-period models. Next, the selected DT was used for the thematic classification of the entire cropland area and mapping was then evaluated applying the confusion matrix method to the independent testing dataset that reported 79% overall accuracy. OCIM detected intra-class variations in most crops attributed to variability from local crop calendars, tree-orchard structures and land management operations. Spectral variables (based on VIs) contributed around 90% to the models, although textural variables were necessary to discriminate between most of the permanent crop-fields (orchards, vineyard, alfalfa and meadow). Features extracted from late-summer imagery contributed around 60% in classification model development, whereas mid-spring and early-summer imagery contributed around 30 and 10%, respectively. The Normalized Difference Vegetation Index (NDVI) was used to identify the main groups of crops based on the presence and vigor of green vegetation within the fields, contributing around 50% to the models. In addition, other VIs based on SWIR bands were also crucial to crop identification because of their potential to detect field properties like moisture, vegetation vigor, non-photosynthetic vegetation and bare soil. The OCIM method was built using interpretable rules based on physical properties of the crops studied and it was successful for object-based feature selection and crop identification.

1. Introduction

Concern about greenhouse gas (GHG) emissions has increased in recent years and it is known that agriculture has been a major source of GHG in the atmosphere (Beach et al., 2008). Essential input data for the modeling of GHG variability in agro-ecosystems include 1) identification of which crop is growing and at which developmental stage on specific parcels, 2) mapping of inter-annual crop rotations, and 3) assessment of soil management practices. Determination of crop distribution is also important for the application of agrarian policy actions related to subsidy payments or implementation of agro-environmental measurements. Agricultural land use information is, therefore, updated routinely in many cropland regions in USA and Europe through farmer communications or ground visits of administrative inspectors to selected fields (Peña-Barragán et al., 2008; South et al., 2004). However, this procedure generates errors and discrepancies in declarations and is time-consuming and expensive. Alternatively, analysis of remotely-sensed imagery could be a more reliable and cost-effective methodology for the monitoring of crop-fields across large areas and could provide consistent temporal records as well (Castillejo-González et al., 2009).

* Corresponding author at: One Shield Av., Davis, CA 95616, United States.
E-mail address: jose.pena@fulbrightmail.org (J.M. Peña-Barragán).

0034-4257/$ – see front matter © 2011 Elsevier Inc. All rights reserved.
doi:10.1016/j.rse.2011.01.009

Many investigations have addressed the topic of crop identification via remote sensing (Congalton et al., 1998; Oetter et al., 2001; Ulaby et al., 1982). Recently, Serra and Pons (2008) presented a methodology for mapping and monitoring temporal signatures of six main Mediterranean crops (winter cereals, rice, maize, alfalfa, fruit trees and other crops) using a hybrid classifier and 36 Landsat images obtained from 2002 to 2005. The authors concluded that a multi-temporal approach is essential to obtain high accuracy and recommended the incorporation of crop phenology in the classification methodology. Simonneaux et al. (2008) also used a time series of eight Landsat images to identify four main classes (bare soils, annual vegetation, trees on bare soil and trees on annual understory) using a decision tree algorithm, although they considered that a detailed typology of crops could not be obtained based only on the analysis of the Normalized Difference Vegetation Index (NDVI) profiles. Other studies used the Moderate Resolution Imaging Spectroradiometer (MODIS) time-series to analyze crop phenological changes and to discriminate general crop types over very large areas, although the spatial resolution of this imagery is only appropriate for parcels larger than 32 ha (Wardlow & Egbert, 2008). In a similar investigation, Wardlow et al. (2007) detected regional intra-class variations due to the crops' different regional planting times that represent sub-classes for each crop. The authors concluded that these variations must be addressed in the development of a large-area crop mapping methodology and recommended the use of non-parametric classifiers (e.g., decision tree), which can handle an information class with multiple sub-classes, for accommodating this intra-class variability.

Discrimination between crops is complex for several reasons. First reason is due to agronomic factors, such as: a) different crop types have similar development patterns and growth calendars (similarity between different crops) and b) crop development schedule and crop patterns can be variable in the same crop type due to farmer decisions, local weather, etc. (variability within the same crop). Second reason can be attributed to technical and economic factors due to limitations on satellite resolution. A good quality/cost relationship and a proper combination of spatial (pixel size several times smaller than parcel size), spectral (data on several regions in the optical spectrum) and temporal (images at several dates of crop development) resolution are required. For example, images from the Terra Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER) are widely used for these reasons (Serra & Pons, 2008). Third reason has been that in the conventional pixel-based methodology, pixels are individually classified according to their digital values, but spatial concepts or contextual information are not incorporated. In this case, the misclassification rate is usually high due to 1) similar spectral characteristics of some agricultural classes, 2) spectral variability of the canopy reflectance and the bare soil background within an agricultural field and 3) the presence of mixed pixels located at the boundary between classes (De Wit & Clevers, 2004).

To overcome the problems due to pixel heterogeneity and crop variability within the field, techniques based on objects have been increasingly implemented in remote-sensed image analysis (Blaschke, 2010). The basic idea of Object-Based Image Analysis (OBIA) is the segmentation of the image and the construction of a hierarchical network of homogenous objects that, in the case of crop classifications, matches to plot boundaries. Once an image is segmented, OBIA simultaneously utilizes several data types such as pixel values, contextual information, object features and neighborhood and hierarchical relationships (Baatz & Schäpe, 2000). In the classification process, all pixels in the entire object are assigned to the same class and thus removing the problem of spectral variability and mixed pixels. To take advantage of the OBIA methodology, the within-field features that characterize each crop need to be identified and evaluated for being used in the subsequent object-based classification procedure.

These features can be based either in spectral properties of the vegetation and the plant canopy or in textural properties of the crop structure or even in a combination of both. Most of these crop characteristics can be directly or indirectly estimated by satellite sensors such as ASTER, which operates in sensitive spectral regions such as the visible and near infrared (VNIR) and the short-wave infrared (SWIR). For instance, the VNIR range has been mainly related to leaf pigments (chlorophyll, carotenoids and anthocyanins) and the SWIR is influenced by water, nitrogen concentration and non-photosynthetic components (lignin, cellulose and starch). Foliage and canopy structure also have a strong effect on the spectral response (Asner, 1998; Colwell, 1974; Pinter et al., 2003). Spectral vegetation indices (VI) have been widely used for analyzing and monitoring temporal and spatial variations of crop characteristics. For example, VI based on the VNIR region, such as the NDVI or the Green Vegetation Index (VIgreen) have been used to monitor vegetation conditions and plant structure, showing different response between crops with closed or open canopies (Gitelson et al., 2002; Jiang et al., 2006). Other indices have been developed to assess the influence of background soil, such as the Soil-Adjusted Vegetation Index (SAVI) and its further refinements (Rondeaux et al., 1996). VI based on the SWIR region such as the Normalized Difference Tillage Index (NDTI) or the Lignin Cellulose Absorption (LCA), among others, have been defined to obtain information about crop residue cover and tillage intensity (Daughtry et al., 2005). Factors such as growth stage, canopy structure, planting patterns and soil background not only have an influence in the reflectance signal captured by the sensor and the derived vegetation indices but also define the structural and contextual attributes (textural pattern) of every crop at a parcel-scale. Thus, making a combined used of crop-stage-related vegetation indices and textural pattern might increase the chance for crop identification in all types of fields.

Among the multiple algorithms applied for image classification, the use of Decision trees (DTs) has increased in recent years because it has certain advantages over other methods. A decision tree is a non-parametric rule-based classifier that works like a “white box” because both its structure (formed by several splits) and its final results (terminal leaves) are easy to interpret. Other advantages are that the DT algorithm can be trained quickly, its execution is rapid, it can manage data on different scales and it can be also used for feature selection/reduction (Pal & Mather, 2003). DTs fit well in the OBIA procedure, since the segmentation phase outputs numerous object-related features and the DT is well suited to sort them and select which best describe every class in the classification tree (Laliberte et al., 2007). Many investigations using DTs have demonstrated successful performance in the use of remotely-sensed data for the analysis of general land cover (Friedl & Brodley, 1997), vegetation cover types (Brown de Colstoun et al., 2003), fire occurrence factors (Lozano et al., 2008), urban classes (Thomas et al., 2003), tropical forest types (Sesnie et al., 2008), and crop yield variability (Zheng et al., 2009); but DTs have not been extensively evaluated in studies with numerous crop types.

In this paper we propose a methodology named Object-based Crop Identification and Mapping (OCIM), which combines OBIA and DTs for the assessment of a large number of major crop types and field status in the agricultural region of Yolo County, California. This county has diversity of cropping systems that are representative of northern and central California agriculture and other Mediterranean climate regions. The objectives were to: 1) explore the relationship of multiple vegetation indices and textural features with the studied crops at parcel scale; 2) evaluate the contribution of the visible, near-infrared and short-wave-infrared spectral regions in the identification of crop types and specific crop management techniques; and 3) determine the influence of different growing-season periods in the crop discrimination.

2. Material and methods

2.1. Description of study area and typical cropping schedule

For this study, we used total cropland area of Yolo County, California (center coordinates 38.5°N, 121.5°W, Datum NAD83). This region has a Mediterranean climate, characterized by hot, dry summers (daily maximum temperatures over 38 °C) and cool winters (daily maximum temperatures of 16 °C). Rainfall ranges from 200 to 460 mm and occurs mostly from November through March, but crops are irrigated during the summer growing season. Soils are mostly deep, as is common in the rich alluvial plains of the Sacramento River. Yolo County has a total agricultural surface of 177,000 ha (out of 264,900 ha of total county area), with a very diverse cropping pattern, representative of the California Central Valley agriculture. Listed in order of total crop area, we studied the major crops of this county: alfalfa and other hays, tomato, winter cereals (mainly wheat, oat, and rye), rice, sunflower, walnut, safflower, vineyard, corn and almond. These crops cover 86% of the cropland surface in Yolo County, and the remaining 14% consist of a combination of other minor crops, up to 130 different commodities. Apart from this area, dry meadows (pastureland and rangeland) represent around 20% of the county surface (Yolo County Agriculture Department, 2006).

We defined the crop calendars of each studied crop through regular field trials conducted in Yolo County, consultation with experts and literature research (California Rice Research Board, 2009; Kaffka & Kearney, 1998; LeBoeuf, 2002; USDA-NASS, 1997, 2006) (Fig. 1). The crop calendar details average information on regular crop stages, although natural variations in planting/sowing and harvesting dates may exist in different years and geographical areas due to several factors related to farmer decisions, weather conditions, etc. Orchards and vineyards are permanent crops that usually show green cover from spring to fall and woody structures during their dormancy period, although their spectral response might be also affected by bare-soil background depending on the plantation pattern (tree size, tree separation, etc.) (UC DANR, 1996, 1998). Alfalfa and hays also permanently cover the soil, although they are highly influenced by the cutting schedule (between six and ten times a year), resulting in high spatial and temporal variability of alfalfa's dense canopy and crown structure within a year (Putnam et al., 2007). On the other hand, non-permanent crops are represented by winter cereal, which grows from mid-fall to late-spring, and summer crops, which grow from mid-spring to late-summer. In this case, crop cover changes from small green plants in early stages (high bare-soil background influence on spectral response), to total green cover in maximum development stages (i.e., no bare-soil influence) and to yellowish or brownish vegetation in the senescent stages at the end of the crop season (i.e., variable bare-soil influence). After harvest, the farmers mostly use two different systems. In conservation systems, crop residue covers the fields in order to protect the soil until the next crop. In traditional systems, tillage operations remove residues and leave the soil surface bare. Thus, we evaluated reflectance and texture values on remote images as affected by different spectral response and planting pattern within crop-fields along crop calendar stages.

2.2. Satellite imagery and derived vegetation indices

Satellite imagery of three different dates were selected in 2006 based on the pre-study of the crop calendars with the need to include key growth stages; the three dates correspond to mid-spring (May 7th), early-summer (June 24th) and late-summer (September 12th) growing-season periods. We selected 2006 because cloud-free images that covered the study zone entirely were available on these key dates, unlike other years in which fewer good-quality images were available. Between two and three ASTER scenes were acquired on each date to completely cover the studied zone. Each image was co-registered and corrected with the software ENVI 4.5 (Research System Inc., Boulder, CO, USA). For our study, we selected the visible (green and red) and near-infrared (NIR) bands with 15 m of spatial resolution and all the six shortwave infrared (SWIR) bands with 30 m of spatial resolution. Both resolutions were combined by resampling the 30-m images to the 15-m pixel size. In this process, the resampling algorithm uses the pixel value of the pixel closest to the center of the source matrix to be resampled. Atmospheric correction and radiometric calibration were performed using Fast Line-of-Site Atmospheric Analysis of Spectral Hypercubes (FLAASH) tool, which is based on MODTRAN4 radiative transfer code (Matthew et al., 2000). Model parameters included U.S. standard atmosphere and rural aerosol

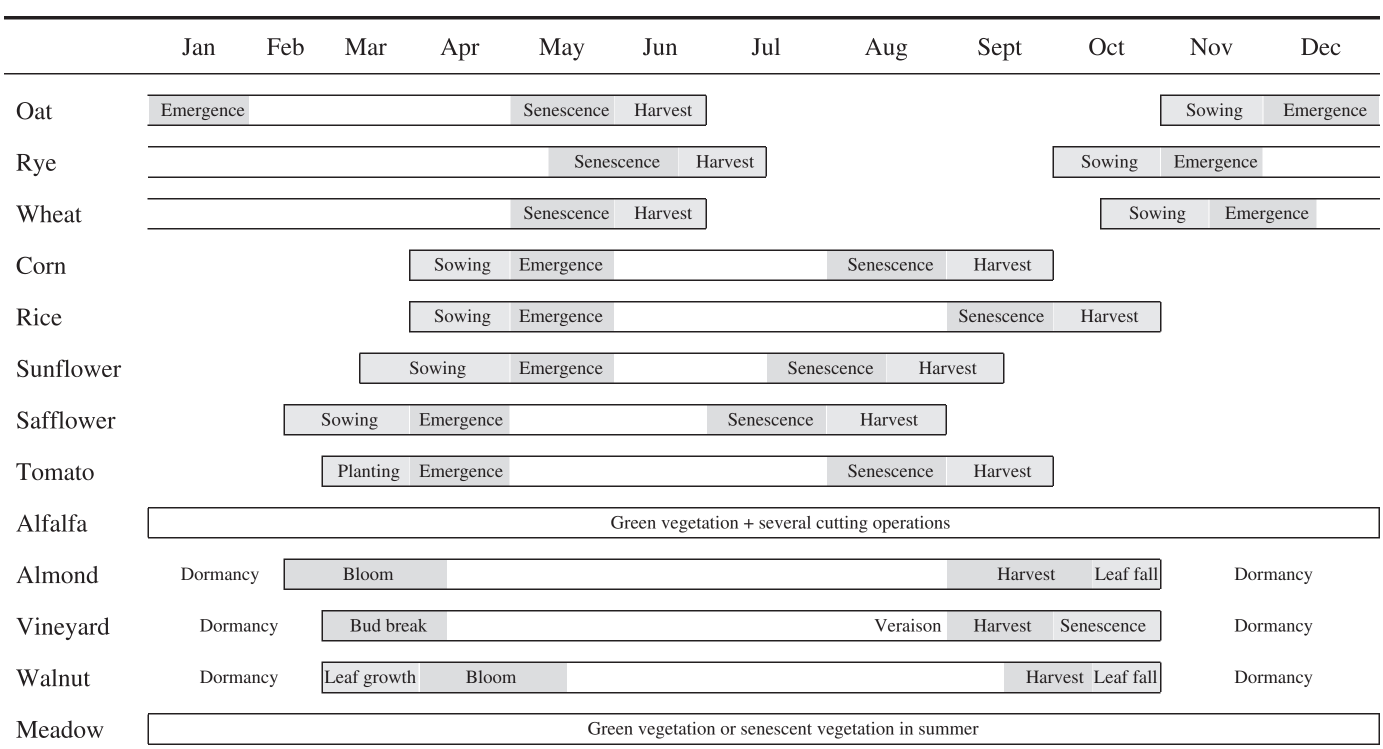

Fig. 1. Crop calendar for major crops growing in Yolo County (CA).

models with visibility set to 40 km. This process involves conversion of the original row digital values to reflectance values.

Twenty vegetation indices (VIs) derived from ASTER wavebands were calculated in the three periods studied (Table 1). These indices have been traditionally used in agricultural studies and they have shown potential to detect certain cropfield characteristic that could be crucial for crop differentiation. We classified VIs according to the spectral region used, as follows: 1) Visible-region indices, 2) Visible–NIR-region indices, 3) NIR–SWIR-region indices, 4) Visible–SWIR-region indices, and 5) SWIR-region indices. The set of VIs was included in the initial pool used to build DT models for crop identification and the results were interpreted according to the crop attributes as affected by the period and the spectrum region studied.

2.3. Object-based Crop Identification and Mapping (OCIM) methodology

The OCIM methodology consists mainly of two consecutive phases: 1) delimitation of crop-fields by image segmentation and, 2) application of decision rules based on physically interpretable thresholds of selected spectral and texture features which characterize every studied crop as affected by growing-season period, crop calendars, and soil management techniques. Next, the methodology used in each phase as well as the statistical analysis used for feature selection and model validation is detailed.

2.3.1. Field delimitation by segmentation

Remote images were segmented into multi-pixel objects using the multiresolution algorithm included in the commercial software Definiens Developer 7 (Definiens AG, Munich, Germany). Segmentation is a bottom–up region-merging process in which the image is subdivided into homogeneous regions according to several parameters (band weights, scale, color, shape, smoothness and compactness) defined by the operator, with the objective of creating objects delimiting crop-field borders. For a complete description of the segmentation algorithm and parameters involved in this process, readers can refer to Definiens manual (Definiens, 2007). In a parallel investigation, we studied all the 1312 different segmentation scenarios made from every possible combination of the parameters involved in this process with the objective of selecting the optimal set of parameters that performed the best segmentation. We applied an empirical discrepancy method (Ortiz & Oliver, 2006; Zhang, 1996) to measure the quality of the segmentation outputs of every scenario compared to 350 ground-truth reference objects that represent fields from all the studied crop types, and assessed the similarity in area, perimeter and shape index between segmented objects and real fields. As a result of this study, values of 0.4, 0.6, 0.2 and 0.8 for scale 50 and 0.9, 0.1, 0.3 and 0.7 for scale 100 were selected for color, shape, smoothness and compactness, respectively, delimiting all the fields with a 94% of averaged accuracy. A segmentation process of scale 50 was initially performed to create "tiny" objects fitted to the field

Table 1
Vegetation indices explored in this study.

Spectral region/vegetation index (VI)	VI adapted to ASTER	Adapted from reference	Commonly related to
Visible region			
Greenness Index (G)	Aster1/Aster2		Leaf pigments, vegetation greenish
Green Vegetation Index (VIgreen)	(Aster1 − Aster2)/(Aster1 + Aster2)	Gitelson et al. (2002)	Leaf pigments, vegetation greenish
Visible–NIR region			
Normalized Difference VI (NDVI)	(Aster3 − Aster2)/(Aster3 + Aster2)	Rouse et al. (1973)	Vegetation status, canopy structure
Green NDVI (GNDVI)	(Aster3 − Aster1)/(Aster3 + Aster1)	Gitelson and Merzlyak (1996)	Vegetation status, canopy structure
Renormalized Difference VI (RDVI)	$\frac{(Aster3 - Aster2)}{\sqrt{(Aster3 + Aster2)}}$	Rougean and Breon (1995)	Vegetation status, canopy structure
Simple Ratio Index (SRI)	Aster3/Aster2	Jordan (1969)	Vegetation status, canopy structure
Modified Simple Ratio (MSR)	$\frac{\left(Aster3 / Aster2\right) - 1}{\left(Aster3 / Aster2\right)^{0.5} + 1}$	Chen (1996)	Vegetation status, canopy structure
Modified Chlorophyll Absorption in Reflectance Index (MCARI)	[(Aster3 − Aster2) − 0.2 (Aster3 − Aster1)] ×(Aster3/Aster2)	Daughtry et al. (2000)	Leaf pigments, vegetation status
Transformed CARI (TCARI)	3 [(Aster3 − Aster2) − 0.2 (Aster3 − Aster1) ×(Aster3/Aster2)]	Haboudane et al. (2002)	Leaf pigments, vegetation status
Triangular Vegetation Index (TVI)	0.5 [120 (Aster3 − Aster1) − 200 (Aster2 − Aster1)]	Broge and Leblanc (2000)	Leaf pigments, vegetation status
NIR–SWIR region			
Normalized Difference Index 5 (NDI5)	(Aster3 − Aster4)/(Aster3 + Aster4)	McNairn and Protz (1993)	Vegetation status, water content, residue cover
Normalized Difference Index for ASTER 3–5 (NDIfA3–5)	(Aster3 − Aster5)/(Aster3 + Aster5)	This study	Vegetation status, water content, residue cover
Normalized Difference Index 7 (NDI7)	(Aster3 − Aster6)/(Aster3 + Aster6)	McNairn and Protz (1993)	Vegetation status, water content, residue cover
Visible–SWIR region			
Normalized Differential Senescent VI (NDSVI)	(Aster4 − Aster2)/(Aster4 + Aster2)	Qi et al. (2002)	Vegetation status, water content, residue cover
Normalized Different Residue Index (NDRI)	(Aster2 − Aster5)/(Aster2 + Aster5)	Gelder et al. (2009)	Vegetation status, water content, residue cover
SWIR region			
Normalized Difference Tillage Index (NDTI)	(Aster4 − Aster6)/(Aster4 + Aster6)	van Deventer et al. (1997)	Non-photosynthetic components, residue cover
Lignin Cellulose Absorption (LCA)	100 [(Aster6 − Aster5) + (Aster6 − Aster8)]	Daughtry et al. (2005)	Non-photosynthetic components, residue cover
ASTER Normalized Shortwave Index (ANSI)	(Aster5 − Aster4)/(Aster5 + Aster4)	Lewis et al. (2006)	Non-photosynthetic components, residue cover
Normalized Difference Index for ASTER 5–6 (NDIfA5–6)	(Aster5 − Aster6)/(Aster5 + Aster6)	This study	Non-photosynthetic components, residue cover
ASTER Shortwave Infrared Normalized Difference Residue Index (SINDRI)	(Aster6 − Aster7)/(Aster6 + Aster7)	Serbin et al. (2009a,b)	Non-photosynthetic components, residue cover

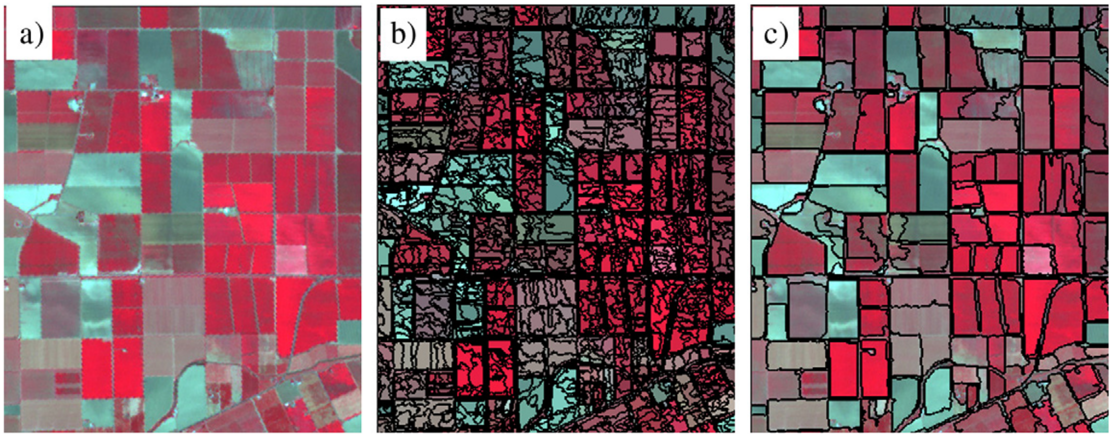

Fig. 2. Partial output images of the Object-based Crop Identification and Mapping (OCIM) methodology: a) portion of the mid-spring ASTER image (15-m spatial resolution) in false-color (G, R and NIR bands) combination, b) segmentation output at the lowest scale to create "tiny" objects fitted to the field borders (level 1, down), and c) segmentation output at the largest scale to create objects similar to the crop fields (level 2, up).

borders (level 1, down) and then a merging process of scale 100 (level 2, top) was carried out to create the smaller fields (5–10 ha). After that, larger fields (>10 ha) were defined by merging neighboring objects with very similar spectral characteristics in the second level (Fig. 2). Thus, a hierarchical network was formed and relationships between sub- and super-objects from the two levels, as well as between adjacent objects (horizontal neighbors) defined new features that might improve within-field crop identification. This large approach was performed for the development of the OCIM methodology and, therefore, the selected parameters could be applied to segment ASTER imagery in other studied areas where average size of the crop parcels was similar to previously mentioned.

2.3.2. Sampling dataset and feature definition

A total of 1300 fields were randomly selected after the segmentation process, representing around 20% of the Yolo County agricultural surface. The number of fields selected for each type of crop was proportional to its total area in the studied zone, from a minimum of 30 fields for minor

Table 2
Description of spectral and texture features used in this study.

Category/name	Equation[a]	# Layers[b]
Object spectral		
Mean	$\frac{1}{\#P_{Obj}} \sum_{(x,y)\in P_{Obj}} c_k(x,y)$	3 Dates × 9 Bands 3 Dates × 20 VIs
Standard deviation	$\sqrt{\frac{1}{\#P_{Obj}} \sum_{(x,y)\in P_{Obj}} \left(c_k(x,y) - \frac{1}{\#P_{Obj}} \sum_{(x,y)\in P_{Obj}} c_k(x,y)\right)^2}$	3 Dates × 9 Bands 3 Dates × 20 VIs
Object texture after Haralick[c]		
GLCM homogeneity	$\sum_{i,j=0}^{N-1} \frac{P_{i,j}}{1+(i-j)^2}$	3 Dates × 3 Bands
GLCM dissimilarity	$\sum_{i,j=0}^{N-1} P_{i,j}\lvert i-j\rvert$	3 Dates × 3 Bands
GLCM entropy	$\sum_{i,j=0}^{N-1} P_{i,j}(-\ln P_{i,j})$	3 Dates × 3 Bands
Object texture based on sub-objects		
Rel_Area_ SubObj (VI<t)	$\frac{1}{\#P_{Obj}} \sum_{Sub\in Obj(VI<t)} \#Sub_{Obj} \cdot \#P_{Sub}$	3 Dates × 10 VIs × 3 Thresholds

[a] Parameters:

$P_{Obj} = \{(x,y):(x,y)\in Obj\}$; set of pixels of an image object.
$\#P_{Obj}$ = total number of pixels contained in P_{Obj}.
$c_k(x,y)$ = image layer value at pixel (x,y); where (x,y) are pixel coordinates.
i = the row number of the co-occurrence matrix.
j = the column number of the co-occurrence matrix.
$P_{i,j}$ = the normalized value in the cell i,j; $P_{i,j} = \frac{V_{i,j}}{\sum_{i,j=0}^{N-1} V_{i,j}}$
$V_{i,j}$ = the value in the cell i,j of the co-occurrence matrix.
N = the number of rows or columns of the co-occurrence matrix.

[b] Description of layers used in each feature:

Dates: mid-spring, early-summer and late-summer.
Object spectral features: All the 20 VIs listed in Table 1, and 9 bands corresponding to visible (2), NIR (1) and SWIR (6) ASTER wavebands.
Object texture after Haralick features: 3 bands corresponding to green, red and NIR ASTER wavebands.
Object texture based on sub-objects features: 10 VIs (VIgreen, NDVI, GNDVI, NDI5, NDI7, NDSVI, NDRI, NDTI, LCA, and SINDRI). Thresholds (t) were defined individually according to DTs modeling. Selected thresholds are shown in the Results and discussion section and in Fig. 3.

[c] The GLCM features Contrast, Angular Second Moment, Mean, Standard Deviation and Correlation were studied but not selected in the final models (Definiens, 2007; Haralick et al., 1973).

crops. The crop planted on every sampled field was confirmed from maps provided by the Yolo agricultural commissioner's office. Some sampled fields were also randomly visited to double-check the reliability of the maps and to take field photographs for crop calendar descriptions. The full dataset were divided to three independent sub-sets to perform the decision tree modeling and its evaluation, as explained in the next section.

As compared to pixel-based methods, the application of object-based approach offers the possibility of evaluating new spectral, textural, contextual and hierarchical features. In our study, we evaluated a feature space formed by 336 object features categorized in three groups as defined in Table 2: 1) object spectral information based on mean and standard deviation (SD) values, 2) object textural information based upon the gray-level co-occurrence matrix (GLCM) proposed by Haralick et al. (1973), and 3) object textural information based on sub-objects derived from the hierarchical structure (Definiens, 2007). In the first group, the mean and SD values were calculated for every VI from the values of all pixels forming an object, showing information related to vegetation status, crop structure, residue content, soil background, etc. The SD value can also indicate the degree of local variability of pixel values within the object. In the second group, the texture features based upon the GLCM were calculated by determining how often a pixel with the intensity (gray-level of the band considered) value *i* occurs in a specific spatial relationship to a pixel with the value *j* (Definiens, 2007). In this group, the features homogeneity, contrast, dissimilarity, entropy, angular 2nd moment, matrix mean, matrix SD and correlation were evaluated. Texture provides supplementary information about object properties that might be helpful for discrimination of heterogeneous crop-fields (Pacifici et al., 2009), although they had a disadvantage of high computational cost. The third group was formed by other type of texture features based on sub-object analysis, in which case we defined a new feature to measure the relative area of sub-objects with pixel values under certain VI thresholds. This new feature may offer a texture measure more comprehensible than GLCM parameters whether both the VI and its physically interpretable thresholds are properly selected. In this group, we selected 10 VIs after a preliminary study made over a reduced area in order to reduce computational time in the subsequent analysis. Other object features based on geometric properties such as shape, perimeter, shape index or density were not considered because they do not have a consistent contribution in the vegetation identification (Ke et al., 2010; Laliberte et al., 2007).

2.3.3. Decision tree modeling and model evaluation

The model design was conducted using a training/validation/testing dataset procedure where, firstly, the training and validation sets were used to create, prune and evaluate the decision trees and, later, the testing set was used to measure the accuracy of the image classification mapping by applying the confusion matrix method (Wright & Gallant, 2007). The size of training and validation set was 1n/4 each one and the remaining 2n/4 was used for the test set, n being the size of the full dataset (n = 1300 fields).

a) DT creation: A pool with all the spectral and texture features defined in Table 2 was used to build DT models for crop identification. We studied four different scenarios, one using the features from the three study dates and the other ones using only two study dates, in order to assess the influence of each growing-season period in the final models. Modeling was performed using the recursive partitioning platform of the statistical software JMP 8 (SAS Institute Inc., Cary, NC, USA). The tree is built by binary recursive splitting of the training set, choosing the feature and the cutting value that best fit the partial response in every split. The algorithm examines a very large number of possible splits and determines the most significant one by the largest likelihood-ratio chi-square (χ^2) statistic, commonly used in testing categorical data. The χ^2 statistic involves the ratios between the observed (O_{ij}) and the expected (E_{ij}) frequencies. In either case, the split is chosen to maximize the difference in the responses between the two branches of the split. This statistic is computed as:

$$\chi^2 = 2\sum_{ij} O_{ij}\, ln\left(\frac{O_{ij}}{E_{ij}}\right).$$

b) Training and validation phase: We applied a cross-validation method for the definition and evaluation of the models, in which sampling fields were randomly separated to 50% for model training and 50% for model validation. This procedure was repeated to generate ten versions of the data with random combinations of training and validation sets (Friedl & Brodley, 1997). We used the measure performance over separate validation dataset to prevent overfitting by post-pruning of the full tree and avoiding splits not statistically significant (Schaffer, 1993; Xu et al., 2005). Because the objective of using DT models is the identification of meaningful features and thresholds for crop discrimination, we looked for similar results in every partial DT. Finally, the best DT was chosen for every scenario by selecting the optimal splits (features and cutting threshold) that yielded the smallest error rate as a whole measured by the proportion of incorrect prediction that the DT makes on the independent validation dataset (Mingers, 1989). DT modeling is an exploratory method for the investigation of relationships interactively. Therefore, decision rules derived from the DT models were interpreted by plotting every node, with the aim of revealing the meaning or physical background of the features and thresholds selected in every split and the information related to regional crop calendars.
c) Mapping evaluation: The selected DT was used to generate a crop-field classification map of the total studied area. The accuracy of the resulting classification was then measured by applying the confusion matrix method to the third independent testing dataset. The coincidence between classified and ground-truth regions was assessed at the field scale by comparing every testing field to its overlapped classified field. User's and producer's accuracy for each crop type and overall accuracy for each group of similar crops and for the general classification were calculated.

3. Results and discussion

3.1. Description of the OCIM methodology by using decision tree modeling

We developed the OCIM methodology using the capacity of the decision tree approach to select the object-based features and their cutting thresholds that best fitted to every type of crop (Fig. 3). Several sub-classes were distinguished for each crop due to intra-crop variations caused mainly by local crop calendars and the application of diverse crop management practices in the studied area (Table 3), as described below.

3.1.1. Identification of general groups of crops

The first phase of the OCIM methodology is based on discriminating between winter cereals, meadows, summer crops and permanent crops, which form the general groups according to crop calendar and growing-season periods. The combination of NDVI calculated in mid-spring and in late-summer was selected in this initial step (Fig. 4). NDVI values lower than 0.25 were associated to fields lacking of green vegetation such as most summer crops in mid-spring (except early-emerging rice fields) and winter cereal and most meadow fields in late-summer. All the permanent crops (vineyard, nut-tree orchard and alfalfa), plus early-emerging rice and some meadow fields, were characterized by NDVI values greater than 0.25 in the three studied periods.

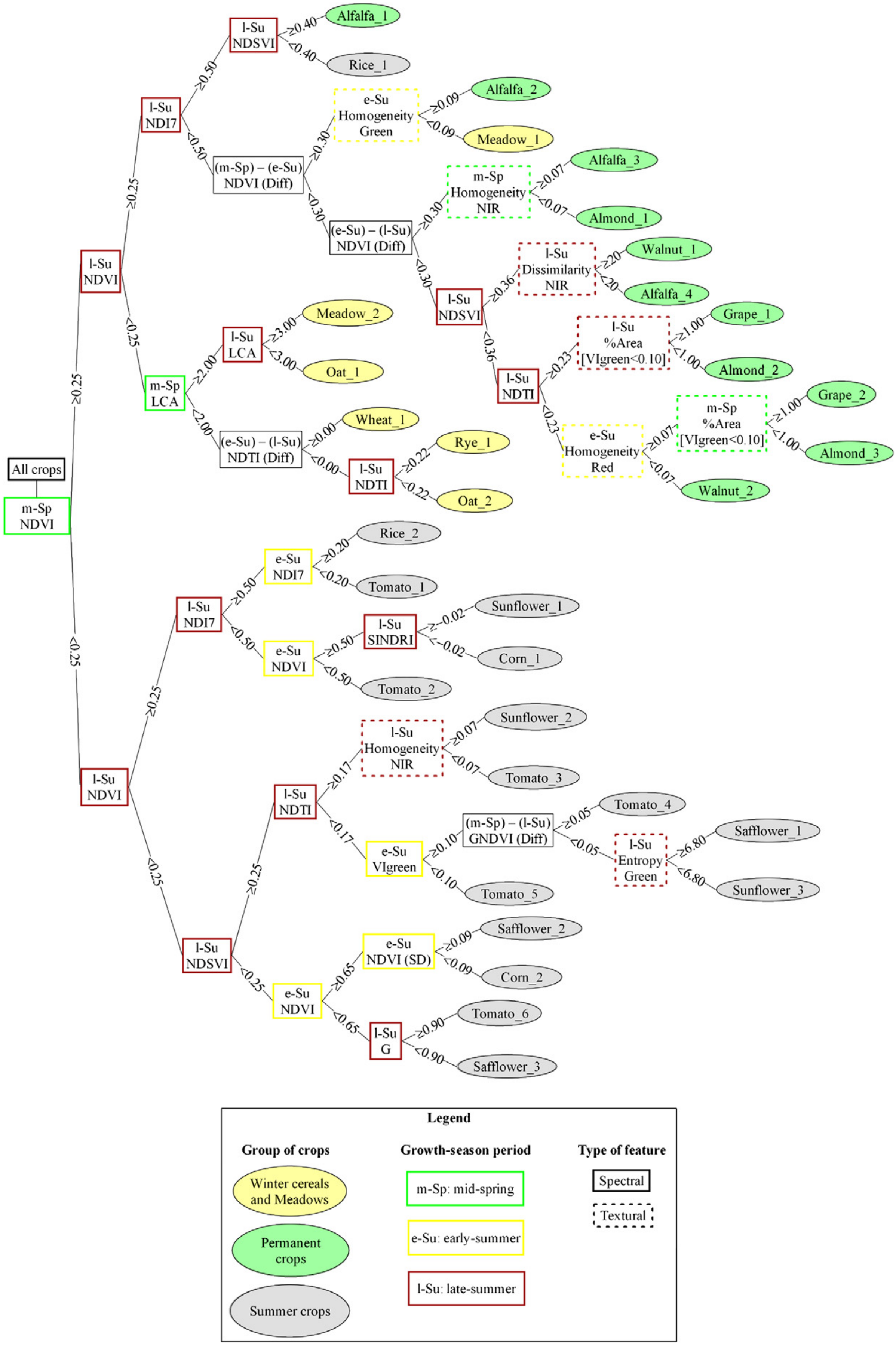

Fig. 3. Decision tree for Object-based Crop Identification and Mapping (OCIM) methodology.

3.1.2. Identification of permanent crops, early-emerging rice and perennial meadow

Permanent crops, plus early-emerging rice and perennial meadow fields, were discriminated by means of a sequence of four VIs (NDVI, NDI7, NDSVI and NDTI) and three texture features (homogeneity, dissimilarity and relative-area-of-subobjects for VIgreen) (Fig. 5). High NDI7 values (>0.50) were strongly characteristic of flooded fields such as all the rice fields and some irrigated alfalfa fields. Many alfalfa fields were also characterized by high changes of NDVI values between consecutive periods (>30%), because of the effect of cutting operations on their greening up and vigor. The remaining permanent crops were then split using the NDSVI index in late-summer, grouping the fields in alfalfa and walnut for values larger than 0.36 and in vineyard, almond and low-density walnut fields for lower values (Fig. 5a). Beyond this split, no spectral index showed capability to improve the discrimination between alfalfa and walnut, but they were efficiently separated by using the dissimilarity texture feature in late-summer (Fig. 5b). In the other group, most of the vineyards were related to NDTI values greater than 0.23 in late-summer, the remaining fields being mostly almonds and low-density walnuts. At this point, mixing up of almond and walnut fields was slightly reduced by assigning low homogeneity texture values (<0.07) to the walnut fields (Fig. 5c) and the identification of vineyards was also enhanced by using a texture feature based on sub-objects (all subobjects with VIgreen values lower than 0.10) in mid-spring and late-summer.

3.1.3. Identification of winter cereals and deciduous meadow

Winter cereal (rye, oat and wheat) and deciduous meadow fields were characterized by high green vegetation vigor in mid-spring and either senescent vegetation, residue or bare soil in late-summer. The LCA index calculated in mid-spring and late-summer was selected by the DT algorithm for the general discrimination between these fields (Fig. 6). High LCA values in both periods were characteristic of meadow fields, high values in mid-spring and low-values in late-summer were typical of some oat fields, while confusion remained in the rest of the fields (Fig. 6a). No spectral or texture feature offered a

Table 3
Description of crop sub-classification defined in the OCIM methodology.

Crop	Sub-class description
Winter cereals and meadow	
Oat	
Oat_1	Greenish vegetation in mid-spring.
Oat_2	Initial desiccation of plants in mid-spring.
Rye	
Rye_1	Greenish vegetation in mid-spring (either residue cover or bare soil after harvest).
Wheat	
Wheat_1	Either greenish vegetation or initial desiccation in mid-spring (either residue cover or bare soil after harvest).
Meadow	
Meadow_1	Perennial meadow: permanent greenish vegetation throughout the year.
Meadow_2	Deciduous meadow: greenish vegetation in mid-spring and desiccated vegetation in late-summer.
Permanent crops	
Alfalfa	
Alfalfa_1	Very high vegetation vigor and high in-field moisture in late-summer.
Alfalfa_2	Vegetation vigor in mid-spring much higher than in early-summer. Cutting operation close to early-summer.
Alfalfa_3	Vegetation vigor in early-summer much higher than in late-summer. Cutting operation close to late-summer.
Alfalfa_4	Remaining alfalfa fields.
Grape	
Grape_1	Most of grape fields. Yellowish vegetation (veraison) in late-summer.
Grape_2	Remaining grape fields. Greenish vegetation in late-summer, related to late development in spring.
Almond	
Almond_1	Vegetation vigor in early-summer much higher than in late-summer, due to advanced leaf fall in late-summer.
Almond_2	Young or small almond trees. Soil background effect greater than in mature orchards.
Almond_3	Most of mature almond orchards. Trees typically spaced to 7–8 m.
Walnut	
Walnut_1	Most of mature walnut orchards. Trees typically spaced to 9 m.
Walnut_2	Young or small walnut trees. Soil background effect greater than in mature orchards.
Summer crops	
Corn	
Corn_1	Greenish to yellowish vegetation in late-summer.
Corn_2	Desiccated vegetation or crop harvested in late-summer (either residue cover or bare soil after harvest).
Rice	
Rice_1	Early-emerging variety. Greenish vegetation in mid-spring.
Rice_2	Most of rice fields. Vegetation still not appreciable in mid-spring.
Safflower	
Safflower_1	High vegetation vigor in early-summer and desiccated vegetation in late-summer, due to late-sowing in spring.
Safflower_2	High vegetation vigor in early-summer and crop harvested in late-summer.
Safflower_3	Most of safflower fields. Moderate vegetation vigor in early-summer and crop harvested in late-summer.
Sunflower	
Sunflower_1	Greenish to yellowish vegetation in late-summer, due to late-sowing in spring.
Sunflower_2	High vegetation vigor in early-summer and desiccated vegetation in late-summer.
Sunflower_3	Crop harvested in late-summer (either residue cover or bare soil after harvest).
Tomato	
Tomato_1	Late-planted fields. Growing from early-summer to early-fall. Greenish vegetation and high in-field moisture in late-summer.
Tomato_2	Late-planted fields. Growing from early-summer to early-fall. Greenish vegetation in late-summer.
Tomato_3	Growing from early-summer to late-summer. Desiccated vegetation or residue cover after harvest in late-summer.
Tomato_4	Growing from early-summer to late-summer. High vegetation vigor in early-summer and desiccated vegetation or crop harvested in late-summer.
Tomato_5	Growing from late-spring to early-summer. Moderate green vegetation in early-summer and crop harvested in late-summer (crop residue remaining).
Tomato_6	Growing from late-spring to early-summer. Moderate green vegetation in early-summer and crop harvested in late-summer.

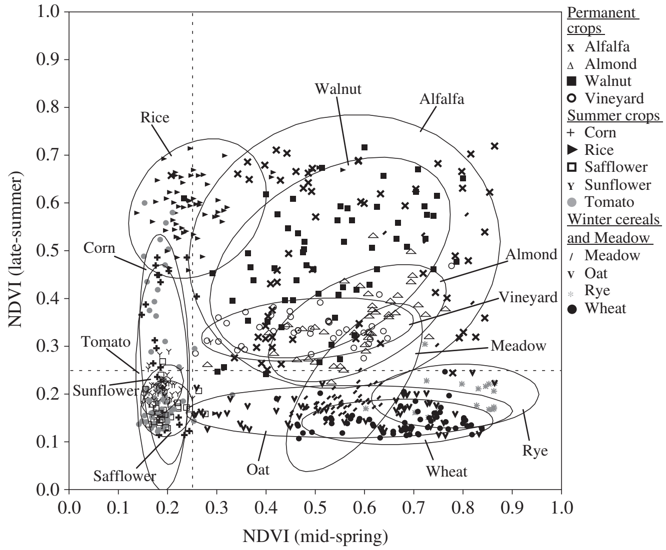

Fig. 4. Relationship between NDVI values in mid-spring versus late-summer and thresholds for discrimination of general groups of crops in the OCIM methodology (ellipses represent 90% of the points of each cloud).

consistent solution for the discrimination between winter cereals because of the similarity between these type of crops, but several VIs based on SWIR bands were useful for the identification of field status after harvest (residue content or bare soil). Comparison of NDTI values in early-summer and late-summer showed that residue fraction generally remained for a longer period in rye fields than in oat or wheat fields (Fig. 6b), because the rye fields entered senescence later in spring, as indicated by their higher NDVI values in mid-spring (Fig. 4). Likewise, negative slope of NDTI distribution in wheat fields can be interpreted as a reduction or loss of residue cover throughout the summer due to natural degradation and land management operations.

3.1.4. Identification of summer crops

Summer crop (corn, rice, safflower, sunflower and tomato) calendars and field status were very diverse in the studied region, increasing the complexity of crop identification. Comparison of NDVI and NDI7 indices in late-summer distinguished several zones (Fig. 7a). On the first hand, some tomato and corn fields and most rice fields that were still growing in late-summer and a few sunflower fields in senescence phase in the late-summer were identified in the zone enclosed by NDVI values greater than 0.25. No safflower fields

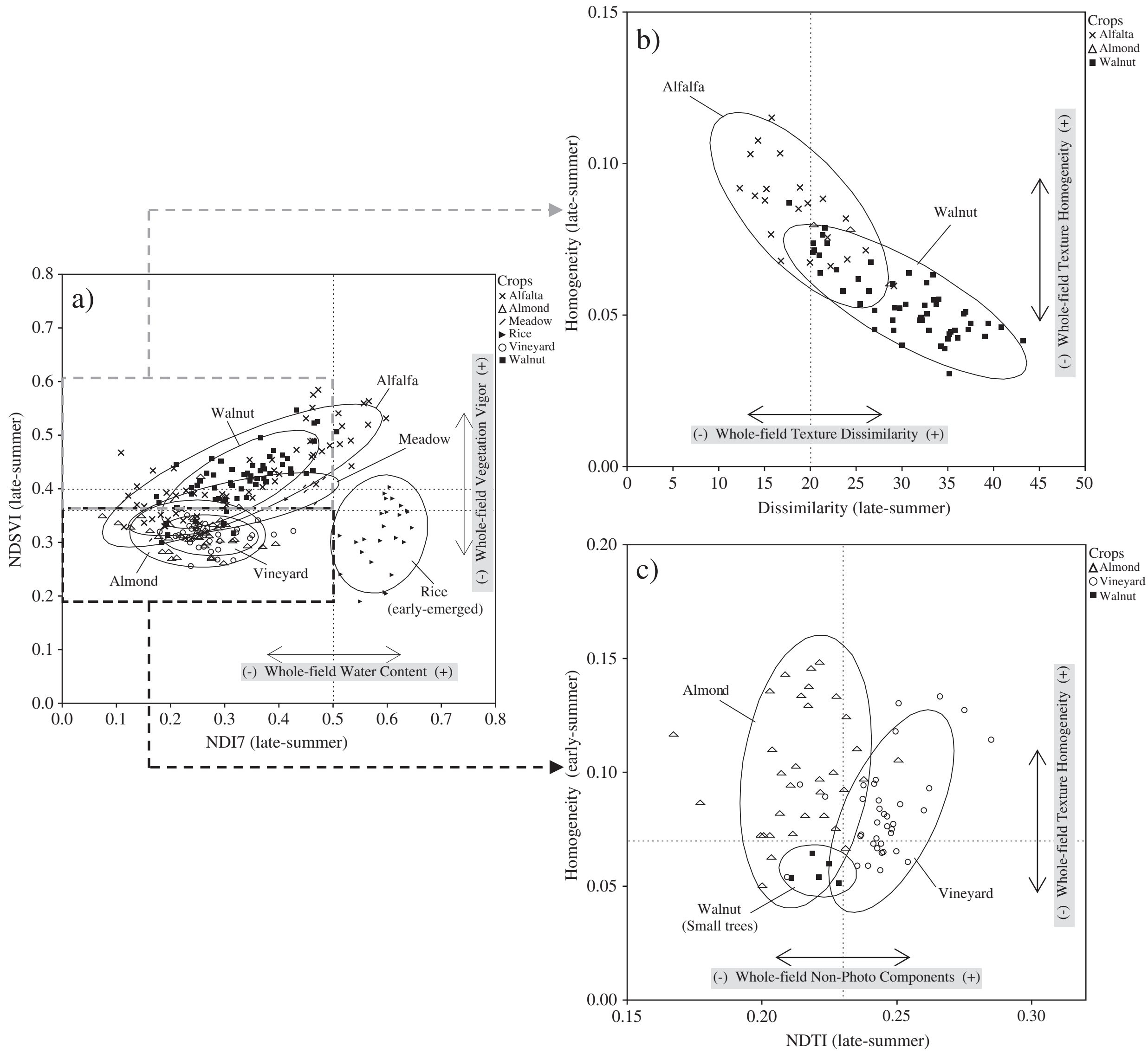

Fig. 5. Relationships between VIs and texture features selected for the identification of early-emerging rice, perennial meadow and permanent crops (vineyard, almond, walnut and alfalfa) in the OCIM methodology (ellipses represent 90% of the points of each cloud).

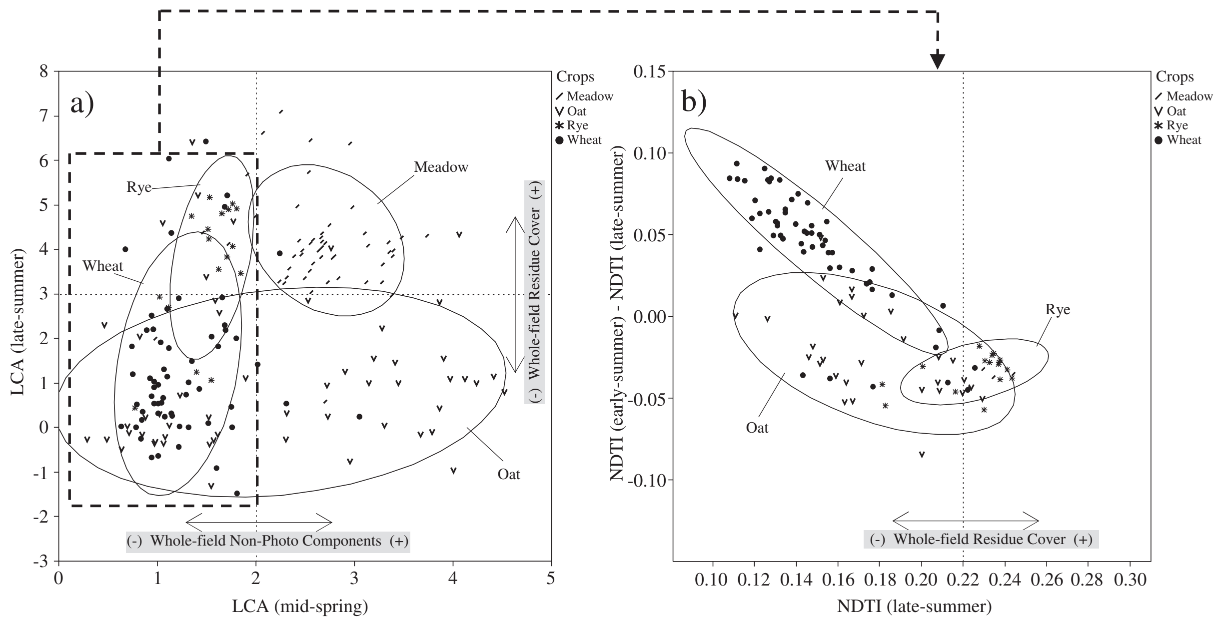

Fig. 6. Relationships between VIs selected for the identification of winter cereals and meadow in the OCIM methodology (ellipses represent 90% of the points of each cloud).

were located in this zone, because this crop was completely desiccated or already harvested in this period. The rice fields were successfully identified by assigning NDI7 values greater than 0.50 in late-summer and than 0.20 in early-summer. NDI7 was the best VI for flooded rice identification because it compares the high reflectance in the NIR band affected by high greenish vigor and the low reflectance in the SWIR region affected by water absorption. On the other hand, confusion between tomato, corn and sunflower fields was solved on a first split by using the NDVI index in early-summer, because the corn and sunflower fields showed greater values (>0.50) than the tomato fields (Fig. 7b). According to the crop calendars, this group of tomato fields were planted later than common (growing period from about early-summer to late-summer or early-fall), resulting in the lower NDVI values in early-summer the higher values in late-summer and vice-versa. The tomato and corn fields that were still growing in late-summer showed similar spectral response in this period, but corn canopy was denser than tomato in early-summer, resulting in greater NDVI values in this period. Uncertainty in the identification of corn versus sunflower fields in late-summer was reduced by using the SINDRI index, which showed higher values for sunflower fields, possibly affected by the advanced senescent stage of these fields and the influence of non-photosynthetic vegetation in the performance of SINDRI (Serbin et al., 2009b).

The remaining fields were characterized by NDVI values lower than 0.25 in late-summer, which is attributed either to very advanced senescent stage, to post-harvest residue cover, or to bare soil in this period. Several VIs related to crop vegetation vigor (mainly NDVI and VIgreen) in early-summer and to crop non-photosynthetic components (NDSVI and NDTI) in late-summer, plus some texture features, were combined to discriminate among these fields. We found that all the remaining tomato and sunflower fields showed NDSVI values higher (>0.25) than all the remaining corn fields and some safflower fields (Fig. 7c). NDSVI index is sensitive to the increase of spectral reflectance in the 1.65-nm SWIR region affected by the manifestation of senescent vegetation due to loss of water in leaf tissues (Qi et al., 2002). Vegetation moisture in both senescent tomato and sunflower plants was generally lower because, according to local crop calendars, these crops were more advanced than in the case of senescent corn in the studied zone. This index also accounts for lower values in fields with bare soil and is affected by soil color and type (Serbin et al., 2009a). Thus, we also suggest that the lower values registered in several safflower and corn fields were possibly due to intense tillage practices in these fields after harvest. After this split, misidentification of some corn versus safflower fields was solved by attributing a greater NDVI value (>0.65) to corn fields in early-summer. On the other hand, NDTI calculated in late-summer performed the best for a general discrimination between the remaining sunflower and tomato fields (Fig. 7d). This index has been used for identification of conventional and conservation tillage practices and, thus, for evaluation of crop residue fraction in several investigations (Daughtry et al., 2005; Gelder et al., 2009; Gowda et al., 2001). Our result reflect that sunflower fields maintain a higher residue content in late-summer than the tomato fields and/or that tillage practices are more intense in tomato than in sunflower fields. Further uncertainty in safflower versus tomato field identification was reduced by using the VIgreen index in early-summer, as most tomato fields showed lower greenish vigor in this period, and between the remaining fields by using the texture features homogeneity and entropy, as sunflower fields showed higher homogeneity and lower entropy in late-summer than tomato and safflower fields, respectively. Both texture features are related either positively (homogeneity) or inversely (entropy) to in-field pixel uniformity, since they measure image homogeneity and image disorder, respectively (Baraldi & Parmiggiani, 1995).

3.2. Feature contribution to the OCIM methodology as affected by feature type, spectral region and growing-season period

The OCIM methodology revealed the spectral and texture features used for the identification of crop fields as affected by the spectral regions and the growing-season periods studied. The importance of each selected feature was defined by its contribution to the total χ^2 statistic and by the number of splits (N) used in the DT (Table 4). On the first hand, the spectral features contributed predominantly (from 86 to 94%) to the DT defined for each studied scenario. The index NDVI represented around 50% of the contribution to the models, and the indices NDI7 and NDSVI between 5 and 15% in all the cases. Although the VIs defined in the SWIR region had, individually, a lower influence than the former VIs, a combination of them (mainly NDTI and LCA) was used between 15 and 25% in the model splits. Discrimination between crops affected by non-photosynthetic vegetation (i.e., senescent stages, residue or woody structures) and bare soil is

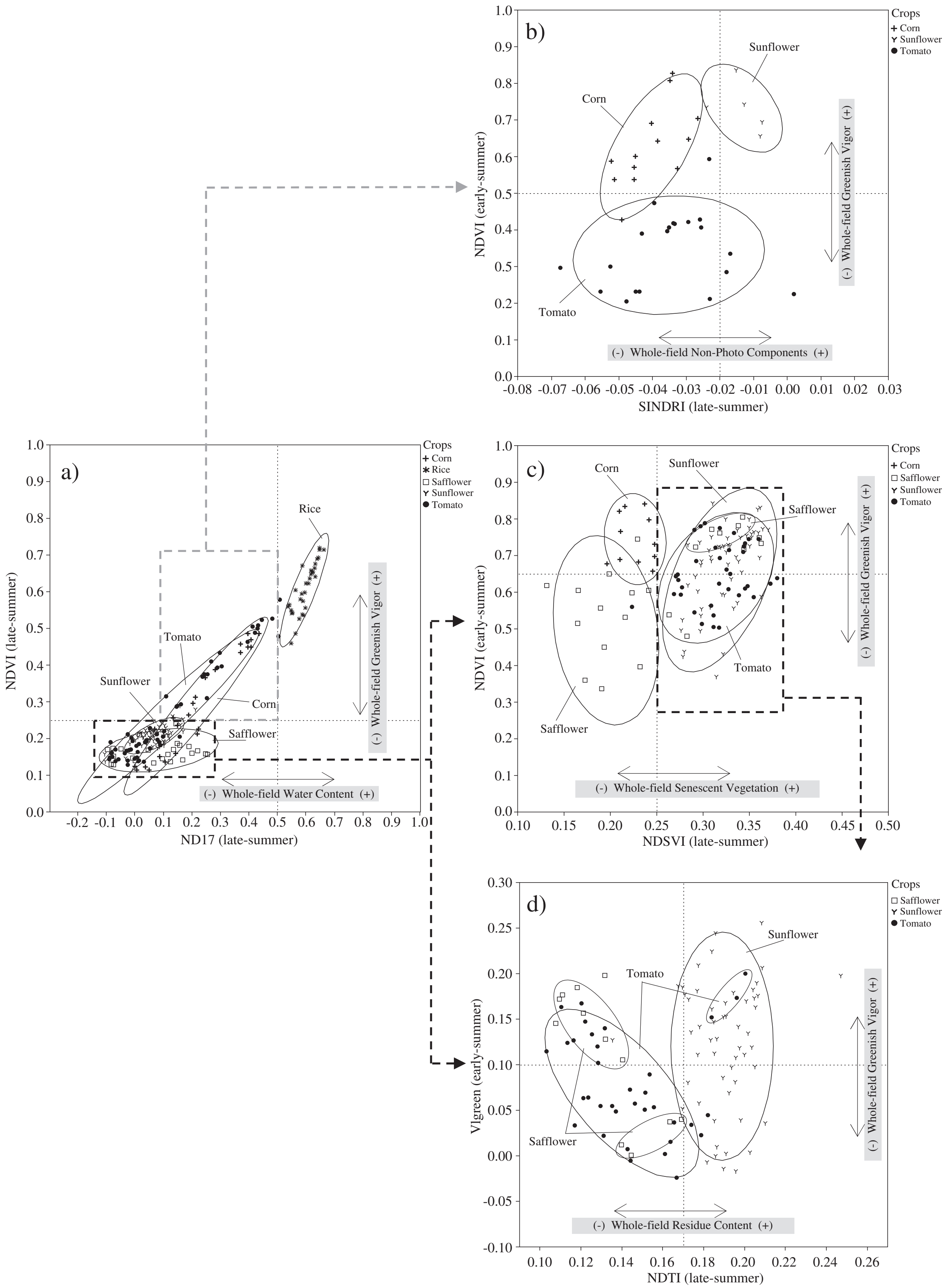

Fig. 7. Relationships between some VIs selected for the identification of summer crops in the OCIM methodology (ellipses represent 90% of the points of each cloud).

Table 4
Contribution of selected VIs and texture features to the OCIM methodology as affected by growing-season period.

Scenario	Spectral region/vegetation index															Texture feature						Total	
	Visible			Visible–NIR			NIR–SWIR			Visible–SWIR			SWIR			Based on sub-objects			Based on Haralick				
	Index	N[a]	$\%\chi^{2b}$	Index	N	$\%\chi^2$	Index	N	$\%\chi^2$	Index	N	$\%\chi^2$	Index	N	$\%\chi^2$	Index	N	$\%\chi^2$	Feature	N	$\%\chi^2$	N	$\%\chi^2$
3 Periods		**2**	**0.77**		**12**	**57.91**		**3**	**10.10**		**3**	**8.17**		**8**	**15.95**		**2**	**2.30**		**6**	**4.79**	**36**	**100.00**
Mid-spring					**3**	**26.04**								**1**	**4.58**		**1**	**0.96**				**5**	**31.58**
				NDVI	2	25.63							LCA	1	4.58	VIgreen	1	0.96					
				GNDVI	1	0.41																	
Early-summer		**1**	**0.68**		**5**	**4.58**		**1**	**1.32**					**1**	**1.20**					**3**	**1.81**	**11**	**9.59**
	VIgreen	1	0.68	NDVI	4	4.15	NDI7	1	1.32				NDTI	1	1.20				Hom-Red	1	0.74		
				SD_NDVI	1	0.43													Hom-NIR	1	0.60		
																			Hom-Green	1	0.47		
Late-summer		**1**	**0.09**		**4**	**27.29**		**2**	**8.78**		**3**	**8.17**		**6**	**10.18**		**1**	**1.34**		**3**	**2.98**	**20**	**58.83**
	G	1	0.09	NDVI	3	26.88	NDI7	2	8.78	NDSVI	3	8.17	NDTI	4	6.34	VIgreen	1	1.34	Dis-NIR	1	1.87		
				GNDVI	1	0.41							LCA	1	3.24				Hom-NIR	1	0.59		
													SINDRI	1	0.60				Ent-Green	1	0.52		
2 Periods[c]		**2**	**1.66**		**7**	**47.04**		**2**	**3.12**		**4**	**10.68**		**13**	**23.90**		**2**	**5.49**		**5**	**8.10**	**35**	**100.00**
Mid-spring		1	0.59		4	26.53					1	5.46		5	13.84					3	4.84	14	51.26
Early-summer		1	1.07		3	20.51		2	3.12		3	5.22		8	10.06		2	5.49		2	3.26	21	48.73
2 Periods		**2**	**0.91**		**8**	**54.80**		**3**	**10.68**		**3**	**8.43**		**10**	**19.60**		**2**	**1.21**		**4**	**4.36**	**32**	**100.00**
Mid-spring		1	0.38		3	24.85		1	1.29					4	7.16		1	0.69		4	4.36	14	38.73
Late-summer		1	0.53		5	29.95		2	9.39		3	8.43		6	12.44		1	0.52				18	61.26
2 Periods		**2**	**2.05**		**4**	**43.77**		**3**	**12.02**		**5**	**14.06**		**12**	**23.14**					**5**	**4.98**	**31**	**100.00**
Early-summer		1	1.48		2	16.47		2	1.53		2	4.38		4	7.34					2	3.36	13	34.56
Late-summer		1	0.57		2	27.30		1	10.49		3	9.68		8	15.80					3	1.62	18	65.46

[a] N: Number of splits used in the DT model.
[b] $\%\chi^2$: Contribution (%) to the total Likelihood-ratio chi-square statistic (χ^2) in every studied scenario.
[c] VIs and features selected in the two-periods scenarios are not shown. Summarized values reflect the contribution of each period, spectral region and feature type to the models.
Bold numbers correspond to sum of values from each period considered.

Table 5
Evaluation of OCIM methodology as affected by growing-season period.

Period	DT validation			Imagery classification assessment (overall accuracy)[a]			
	N[b]	χ^{2}[c]	Error rate (%)	Permanent crops[d] (%)	Summer crops[d] (%)	Winter cereals and meadows[d] (%)	All crops (%)
Mid-spring, early-summer, late-summer	36	1283.60	9	76	84	75	79
Mid-spring, early-summer	35	1097.93	16	67	79	72	64
Mid-spring, late-summer	32	1155.72	13	74	66	68	67
Early-summer, late-summer	31	1230.94	12	68	82	67	73

[a] Based on the confusion matrix method calculated from 650 independent test fields (Table 6). The overall accuracy of permanent crops, summer crops and winter cereals and meadows shows the accuracy among the specific fields of each group of crops.
[b] Number of splits used in the DT model.
[c] Likelihood-ratio chi-square statistic.
[d] Permanent crops: alfalfa, vineyard, almond and walnut; summer crops: corn, rice, safflower, sunflower and tomato; winter cereals: oat, rye and wheat.

difficult in the VNIR spectral region (Nagler et al., 2000), but selected VIs calculated in the SWIR region were sensitive to crop variability at these stages and improved the performance for crop identification. Out of the 20 VIs included in the feature pool, the indices RDVI, SRI, MSR, TCARI, TVI and NDIfA3-5 were selected in none of the scenarios, and the indices MCARI, ANSI, NDIfA5-6 and NDI5 only contributed to minor splits in the two-period scenarios.

On the other hand, the texture features contributed only from 6 to 14% to the models, depending on the scenario considered. These features were secondary variables but important to reduce uncertainty in much of the last splits in the models, mainly between heterogeneous permanent crops. Those features based on Haralick parameters, mainly GLCM homogeneity and dissimilarity, were more used in the models than those based on sub-objects, with the exception of the scenario lacking the late-summer period. Out of the 8 Haralick's features evaluated, the parameters contrast, angular second moment, correlation, mean and standard deviation were selected in none of the scenarios. These statistics provide information about the local variability of the pixel values within a field, but a detailed interpretation is complex. Therefore, future analysis is recommended to determine the relationships between textural properties and certain crop parameters at the field scale, also affected by object heterogeneity or image spatial resolution (Chen et al., 2004).

The features defined in late-summer were the most used both in three- and in two-period scenarios; namely around 60% of the splits were based on the late-summer features. Considering the three-period scenario, the features calculated in mid-spring contributed to around 30% and those defined in early-summer contributed only 10% to the overall χ^2 of the model. Nevertheless, these latter features were used 11 times out of 36, revealing a higher importance to refine the last splits of the DT. Considering the two-period scenarios, the influence of VIs calculated in the SWIR region was considerably higher than for three-period scenarios. The highest influence of the texture features was in the scenario lacking the late-summer period, just the period in which the spectral features contributed the most.

3.3. Evaluation of the OCIM methodology

The OCIM methodology was developed in four different scenarios (combinations of three or two periods) and evaluated by using three independent ground-truth dataset and two consecutive validation methods (Table 5): 1) cross-validation of the DT models, and 2) accuracy assessment of the imagery classification mapping based on the confusion matrix (Table 6). On the first part, the error rate for the three-period DT was 9%, increasing to 12, 13 and 16% in the scenarios lacking the images at mid-spring, early-summer and late-summer, respectively. On the second part, the results of the imagery classification assessment followed a similar order and the overall accuracies in the classification of all the crops were 79% by using the three-period approach and 73%, 67% and 64% using the two-period approaches. The lack of late-summer period in the model yielded the lowest accuracy, demonstrating that this period was decisive for the success of the general crop identification. This result is explained because intra-class variability was higher at the end of summer than in the earlier periods which increased the number of features utilized in this period for the crop identification.

Table 6
Field-based confusion matrix of the OCIM methodology for three-season periods.

User classes	Ground-truth classes														
	Permanent crops				Summer crops					Winter cereals and meadows				Total fields	User's accuracy (%)
	Afalfa	Almond	Walnut	Vineyard	Corn	Rice	Safflower	Sunflower	Tomato	Meadow	Oat	Rye	Wheat		
Afalfa	**48**	5	7	1		1				2		3		67	72
Almond		**24**	4	1										29	83
Walnut	9	2	**44**	2				1		1				59	75
Vineyard	3	7	1	**36**	1									48	75
Corn					**25**	2	2		1					30	83
Rice					1	**57**			3					61	93
Safflower							**22**	4	5					31	71
Sunflower					1		1	**51**	4					57	89
Tomato					2		4	4	**47**					57	82
Meadow		2	1							**51**	10	6	4	74	69
Oat			3							2	45	4	10	64	70
Rye												15		15	100
Wheat							1			4	5	2	**46**	58	79
Total fields	60	40	60	40	30	60	30	60	60	60	60	30	60	650	
Producer's accuracy (%)	80	60	73	90	83	95	73	85	78	85	75	50	77		

Overall producer accuracy = 77%; overall user accuracy = 80%; overall accuracy = 79%; kappa coefficient = 0.75.
Bold values correspond to number of fields correctly classified.

Additionally, Table 5 also quantifies specific classification accuracy of the three general groups of crops in case of lacking any of the studied periods, reporting that permanent crops (alfalfa, vineyard, almond and walnut) yielded the best classification by using mid-spring and late-summer imagery, summer crops (corn, rice, safflower, sunflower and tomato) by using early-summer and late-summer imagery and winter cereals (oat, rye and wheat) and meadows by using mid-spring and early-summer imagery. These results reflect the dates in which each crop is more feasible to be discriminated according to regional crop calendar and are helpful to optimize the imagery collection schedule. For example, discrimination between permanent crops was facilitated by the fact that differences of crop status were larger between late-summer and mid-spring than for early-summer. Although these fields were generally in typical green vegetation in the three studied periods, certain crop stages are more noticeable in the two selected dates. In late-summer, nut-trees were in maximum development but close to leaf fall, whereas grapevines are either in greenish or in yellowish stages depending on their veraison development and alfalfa is drier than in mid-spring and its vegetation vigor is variable depending on irrigation and cutting schedule. In the case of summer crops, early summer was obviously the key date for the discrimination, in addition to the late-summer period when field status generally varies among initial desiccation, senescent vegetation and residue or bare soil in post-harvest, which increase the chance for crop characterization, on contrary to the mid-spring period when these fields are generally bare soil or scarcely covered by the crop. For the task of mapping winter cereal and meadow fields, the mid-spring is crucial for being the moment in which vegetation is turning from greenish to yellowish affected by its desiccation phase. Summer periods had minor influence in this group, although spectral differences between some cereal (soil covered by residue or bare after harvest) and meadow (soil still remained covered by senescent vegetation) fields were detected in this season by using SWIR-based vegetation indices that are sensitive to non-photosynthetic vegetation.

The confusion matrix also reported the partial accuracies for every crop quantified in 650 independent testing fields (Table 6). The crop rice yielded the best result with producer's and user's accuracy of over 93%. This efficiency can be attributed to the fact that this crop growth in flooded fields and its spectral properties are very distinguishable due to effect of water in the NIR and SWIR spectral regions (Sims & Gamon, 2003). The crops alfalfa, vineyard, corn, sunflower and meadow yielded producer's accuracies over 80%, which denotes a good identification performance. Nevertheless, user's accuracy of alfalfa, vineyard and meadow was 72%, 75% and 69%, respectively, which demonstrates that the OCIM methodology slightly overestimated the classification of these crops. The crops almond and rye yielded the lowest producer's accuracy (60% and 50%, respectively). The former was mainly confused with vineyard fields which can be explained by the similar spectral properties and orchard pattern in the cases of almond fields characterized by small trees and somewhat affected by soil background effect. The latter was mainly confused to deciduous meadow with very similar spectral response. Among permanent crops, the major confusion was between alfalfa fields and some lush walnut orchards characterized by high vegetation density, close tree distribution and permanent cover crop between trees. Among summer crops, the major confusion was between some safflower, sunflower and tomato fields with similar crop development according to local calendars. In the case of winter cereals, producer's and user's accuracy was relatively moderate (between 50 and 79%) because only the mid-spring image coincided with their growing stage and, therefore, using more images taken in earlier crop stages might be needed to improve this classification.

4. Conclusions

The Object-based Crop Identification and Mapping (OCIM) methodology developed in this paper was suitable for object-based feature selection and crop identification at different phenological stages and field conditions in Yolo County agroecosystem. The OCIM methodology involves several consecutive steps and provides three levels of crop identification. After segmentation of remote images into realistic and homogeneous areas, cropland was firstly discriminated between permanent crops, summer crops, winter cereals and meadows. Afterwards, the model defines which crop is growing in every parcel and, simultaneously, the intra-class variations attributed to specific crop management operations.

Several spectral (vegetation indices), textural and hierarchical features from three different crop-growing periods were selected in a decision tree structure, revealing the basis of interpretation or physical background affected by every type of crop, as well as their relationships to field crop stages and crop calendars. The spectral features formed the decision tree framework and contributed around 90% to the model. Some textural features were also necessary to discriminate between crop-fields with similar spectral responses, mainly permanent crops (orchards, vineyard, alfalfa and meadow). Out of 336 object-based features studied, the three-period model included 24 features that made relevant contributions, divided them up into 16 spectral, 6 textural and 2 hierarchical features, demonstrating the potential of this technique to optimize the feature selection for crop classification and, consequently, reducing the computational time of image analysis procedure.

NDVI was the index that made the main contribution to OCIM by identifying the principal groups of crops based on the presence and vigor of green vegetation within the fields. In addition, other vegetation indices incorporating SWIR bands (i.e., NDSVI, NDI7, NDTI and LCA) were also crucial to crop identification performance because of their relationship to field properties such as moisture, vegetation vigor, non-photosynthetic vegetation and bare soil. Studying the evolution of crop and field status by using several images in distinct growing stages was essential for crop identification. The features calculated in the late-summer period contributed around 60% to the models produced, followed by mid-spring (30%) and early-summer (10%). Intra-class variations were detected and attributed to different local crop calendars (mainly, planting schedule), orchard structure (i.e., tree size or separation) and land management operations (i.e., cover crops, residue or tillage after harvest).

OCIM was built using interpretable rules based on physical properties of the studied crops and, therefore, most of these rules can be extrapolated to other Mediterranean climate regions, although future analyses are necessary to evaluate its robustness. Moreover, ASTER spatial, spectral and temporal resolution fulfilled requirements regarding general field size and crop calendars and, therefore, similar results are expected using other calibrated multi- or hyper-spectral imagery with similar resolutions such as Landsat or Hyperion.

In a subsequent companion investigation, we are developing an extended rule-set version of OCIM methodology for cropland classification of multi-year satellite imagery, in order to monitor specific crop parameters at a field scale for the integration of remote sensing and modeling to assess temporal and spatial variability of greenhouse gas emissions in agro-ecosystems.

Acknowledgements

We thank the Kearney Foundation of Soil Science for funding this project. Laurie Schuler-Flynn of Yolo County Agriculture Dept kindly provided us ground truth data on crops planted during the study period and we thank her. ASTER data was available to us through NASA EOS scientific investigator affiliation. The research of J.M. Peña-Barragan was granted by the Fulbright-MEC postdoctoral program, financed by the Secretariat of State for Research of the Spanish Ministry for Science and Innovation.

References

Asner, G. P. (1998). Biophysical and biochemical sources of variability in canopy reflectance. *Remote Sensing of Environment*, *64*, 234−253.

Baatz, M., & Schäpe, A. (2000). Multi-resolution segmentation — An optimization approach for high quality multi-scale segmentation. In J. Strbl (Ed.), *Beiträge zum AGIT Symposium Salsburg 2000. (Hrsg.): Angewandte Geographische Informationsverarbeitung, XII*. (pp. 12−23) Karlruhe: Herbert Wichmann Verlag.

Baraldi, A., & Parmiggiani, F. (1995). An investigation of the textural characteristics associated with gray level cooccurrence matrix statistical parameters. *IEEE Transactions on Geoscience and Remote Sensing*, *33*(2), 293−304.

Beach, R. H., DeAngelo, B. J., Rose, S., Li, C., Salas, W., & DelGrosso, S. J. (2008). Mitigation potential and costs for global agricultural greenhouse gas emissions. *Agricultural Economics*, *38*, 109−115.

Blaschke, T. (2010). Object based image analysis for remote sensing. *ISPRS Journal of Photogrammetry and Remote Sensing*, *65*(1), 2−16.

Broge, N. H., & Leblanc, E. (2000). Comparing prediction power and stability of broadband and hyperspectral vegetation indices for estimation of green leaf area index and canopy chlorophyll density. *Remote Sensing of Environment*, *76*(2), 156−172.

Brown De Colstoun, E. C. B., Story, M. H., Thompson, C., Commisso, K., Smith, T. G., & Irons, J. R. (2003). National Park vegetation mapping using multi-temporal Landsat 7 data and a decision tree classifier. *Remote Sensing of Environment*, *85*, 316−327.

California Rice Research Board (2009). http://www.carrb.com (last accessed October 3, 2010).

Castillejo-González, I. L., López-Granados, F., García-Ferrer, A., Peña-Barragán, J. M., Jurado-Expósito, M., de la Orden, M. S., et al. (2009). Object- and pixel-based analysis for mapping crops and their agro-environmental associated measures using QuickBird imagery. *Computers and Electronics in Agriculture*, *68*(2), 207−215.

Chen, J. (1996). Evaluation of vegetation indices and modified simple ratio for boreal applications. *Canadian Journal of Remote Sensing*, *22*, 229−242.

Chen, D., Stow, D. A., & Gong, P. (2004). Examining the effect of spatial resolution and texture window size on classification accuracy: An urban environment case. *International Journal of Remote Sensing*, *25*(11), 2177.

Colwell, J. (1974). Vegetation canopy reflectance. *Remote Sensing of Environment*, *3*, 175−183.

Congalton, R. G., Balogh, M., Bell, C., Green, K., Milliken, J. A., & Ottman, R. (1998). Mapping and monitoring agricultural crops and other land cover in the Lower Colorado River basin. *Photogrammetric Engineering and Remote Sensing*, *64*(11), 1107−1113.

Daughtry, C. S. T., Hunt, E. R., Doraiswamy, P. C., & McMurtrey, J. E. (2005). Remote sensing the spatial distribution of crop residues. *Agronomy Journal*, *97*(3), 864−871.

Daughtry, C. S. T., Walthall, C. L., Kim, M. S., Brown de Colstoun, E., & McMurtrey, J. E., III (2000). Estimating corn leaf chlorophyll concentration from leaf and canopy reflectance. *Remote Sensing of Environment*, *74*, 229−239.

De Wit, A. J. W., & Clevers, J. G. P. W. (2004). Efficiency and accuracy of per-field classification for operational crop mapping. *International Journal of Remote Sensing*, *25*(20), 4091−4112.

Definiens (2007). *Definiens Developer 7: Reference book*. Munich: Definiens AG.

Friedl, M., & Brodley, C. (1997). Decision tree classification of land cover from remotely sensed data. *Remote Sensing of Environment*, *61*(3), 399−409.

Gelder, B. K., Kaleita, A. L., & Cruse, R. M. (2009). Estimating mean field residue cover on Midwestern soils using satellite imagery. *Agronomy Journal*, *101*(3), 635−643.

Gitelson, A. A., Kaufman, Y. J., Stark, R., & Rundquist, D. (2002). Novel algorithms for remote estimation of vegetation fraction. *Remote Sensing of Environment*, *80*(1), 76−87.

Gitelson, A., & Merzlyak, M. N. (1996). Signature analysis of leaf reflectance spectra: Algorithm development for remote sensing of chlorophyll. *Journal of Plant Physiology*, *148*, 494−500.

Gowda, P. H., Dalzell, B. J., Mulla, D. J., & Kollman, F. (2001). Mapping tillage practices with Landsat Thematic Mapper based logistic regression models. *Journal of Soil and Water Conservation*, *56*(2), 91−96.

Haboudane, D., Miller, J. R., Tremblay, N., Zarco-Tejada, P. J., & Dextraze, L. (2002). Integration of hyperspectral vegetation indices for prediction of crop chlorophyll content for application to precision agriculture. *Remote Sensing of Environment*, *81*(2–3), 416−426.

Haralick, R., Shanmugan, K., & Dinstein, I. (1973). Textural features for image classification. *IEEE Transactions on Systems, Man, and Cybernetics*, *3*(1), 610−621.

Jiang, Z., Huete, A. R., Chen, J., Chen, Y., Li, J., Yan, G., et al. (2006). Analysis of NDVI and scaled difference vegetation index retrievals of vegetation fraction. *Remote Sensing of Environment*, *101*(3), 366−378.

Jordan, C. F. (1969). Derivation of leaf area index from quality of light on the forest floor. *Ecology*, *50*, 663−666.

Kaffka, S. R., & Kearney, T. E. (Eds.). (1998). *Safflower production in California. UC agriculture & natural resources publication*, *21565*, .

Ke, Y., Quackenbush, L. J., & Im, J. (2010). Synergistic use of QuickBird multispectral imagery and LIDAR data for object-based forest species classification. *Remote Sensing of Environment*, *114*(6), 1141−1154.

Laliberte, A. S., Fredrickson, E. L., & Rango, A. (2007). Combining decision trees with hierarchical object-oriented image analysis for mapping arid rangelands. *Journal of Photogrammetric Engineering and Remote Sensing*, *73*, 197−207.

LeBoeuf, J. (2002). Crop time line for fresh market tomatoes in California. *Prepared for the U.S. Environmental Protection Agency, Office of Pesticide Programs, Washington, D.C.*

Lewis, D., Yao, H., Fridgen, J., & Kincaid, R. (2006). Investigation on the potential of using ASTER image for corn plant residue coverage estimation in three Indiana counties. *Proceedings of the Geoscience and Remote Sensing Symposium (IGARSS 2006), Denver, CO* (pp. 2092−2094).

Lozano, F. J., Suarez-Seoane, S., Kelly, M., & Luis, E. (2008). A multi-scale approach for modeling fire occurrence probability using satellite data and classification trees: A case study in a mountainous Mediterranean region. *Remote Sensing of Environment*, *112*(3), 708−719.

Matthew, M. W., Adler-Golden, S. M., Berk, A., Richtsmeier, S. C., Levine, R. Y., Bernstein, L. S., et al. (2000). Status of atmospheric correction using a MODTRAN4-based algorithm. *SPIE proceeding. Algorithms for multispectral, hyperspectral, and ultraspectral imagery VI*, *4049*. (pp. 199−207).

McNairn, H., & Protz, R. (1993). Mapping corn residues cover on agricultural fields in Oxford County, Ontario, using thematic mapper. *Canadian Journal of Remote Sensing*, *19*(2), 152−159.

Mingers, J. (1989). An empirical comparison of pruning methods for decision tree induction. *Machine Learning*, *4*(2), 227−243.

Nagler, P. L., Daughtry, C. S. T., & Goward, S. N. (2000). Plant litter and soil reflectance. *Remote Sensing of Environment*, *71*(2), 207−215.

Oetter, D. R., Cohen, W. B., Berterretche, M., Maiersperger, T. K., & Kennedy, R. E. (2001). Land cover mapping in an agricultural setting using multiseasonal Thematic Mapper data. *Remote Sensing of Environment*, *76*(2), 139−155.

Ortiz, A., & Oliver, G. (2006). On the use of the overlapping area matrix for image segmentation evaluation: A survey and new performance measures. *Pattern Recognition Letters*, *27*(16), 1916−1926.

Pacifici, F., Chini, M., & Emery, W. J. (2009). A neural network approach using multi-scale textural metrics from very high-resolution panchromatic imagery for urban land-use classification. *Remote Sensing of Environment*, *113*(6), 1276−1292.

Pal, M., & Mather, P. M. (2003). An assessment of the effectiveness of decision tree methods for land cover classification. *Remote Sensing of Environment*, *86*(4), 554−565.

Peña-Barragán, J. M., López-Granados, F., García Torres, L., Jurado-Expósito, M., Sánchez-de la Orden, M., & García-Ferrer, A. (2008). Discriminating cropping systems and agro-environmental measures by remote sensing. *Agronomy for Sustainable Development*, *28*(2), 355−362.

Pinter, P. J., Hatfield, J. L., Schepers, J. S., Barnes, E. M., Moran, M. S., Daughtry, C. S. T., et al. (2003). Remote sensing for crop management. *Photogrammetric Engineering and Remote Sensing*, *69*(6), 647−664.

Putnam, D. H., Summers, C. G., & Orloff, S. B. (2007). Alfalfa production systems in California. Chapter 1. In C. G. Summers, & D. H. Putnam (Eds.), *Irrigated alfalfa management for Mediterranean and desert zones. Oakland: University of California Agriculture and Natural Resources Publication*, *8287*, See: http://alfalfa.ucdavis.edu/IrrigatedAlfalfa

Qi, J., Marsett, R., Heilman, P., Biedenbender, S., Moran, S., Goodrich, D., et al. (2002). RANGES improves satellite-based information and land cover assessments in southwest United States. *EOS. Transactions of the American Geophysical Union*, *83*, 601−606.

Rondeaux, G., Steven, M., & Baret, F. (1996). Optimization of soil-adjusted vegetation indices. *Remote Sensing of Environment*, *55*(2), 95−107.

Rougean, J. -L., & Breon, F. M. (1995). Estimating PAR absorbed by vegetation from bidirectional reflectance measurements. *Remote Sensing of Environment*, *51*, 375−384.

Rouse, J. W., Haas, R. H., Schell, J. A., & Deering, D. W. (1973). Monitoring vegetation systems in the Great Plains with ERTS. *Proceedings of the Earth Resources Technology Satellite Symposium NASA SP-351, Washington, DC, USA*, *Vol. 1*. (pp. 309−317).

Schaffer, C. (1993). Overfitting avoidance as bias. *Machine Learning*, *10*(2), 153−178.

Serbin, G., Daughtry, C. S. T., Hunt, E. R., Brown, D. J., & McCarty, G. W. (2009). Effect of soil spectral properties on remote sensing of crop residue cover. *Soil Science Society of America Journal*, *73*(5), 1545−1558.

Serbin, G., Hunt, E., Jr., Daughtry, C., McCarty, G., & Doraiswamy, P. (2009). An improved ASTER index for remote sensing of crop residue — MDPI. *Remote Sensing*, *1*(4), 971−991.

Serra, P., & Pons, X. (2008). Monitoring farmers' decisions on Mediterranean irrigated crops using satellite image time series. *International Journal of Remote Sensing*, *29*(8), 2293−2316.

Sesnie, S. E., Gessler, P. E., Finegan, B., & Thessler, S. (2008). Integrating Landsat TM and SRTM-DEM derived variables with decision trees for habitat classification and change detection in complex neotropical environments. *Remote Sensing of Environment*, *112*(5), 2145−2159.

Simonneaux, V., Duchemin, B., Helson, D., Er-Raki, S., Olioso, A., & Chehbouni, A. G. (2008). The use of high-resolution image time series for crop classification and evapotranspiration estimate over an irrigated area in central Morocco. *International Journal of Remote Sensing*, *29*(1), 95.

Sims, D. A., & Gamon, J. A. (2003). Estimation of vegetation water content and photosynthetic tissue area from spectral reflectance: A comparison of indices based on liquid water and chlorophyll absorption features. *Remote Sensing of Environment*, *84*(4), 526−537.

South, S., Qi, J., & Lusch, D. P. (2004). Optimal classification methods for mapping agricultural tillage practices. *Remote Sensing of Environment*, *91*(1), 90−97.

Thomas, N., Hendrix, C., & Congalton, R. G. (2003). A comparison of urban mapping methods using high-resolution digital imagery. *Photogrammetric Engineering and Remote Sensing*, *69*(9), 963−972.

UC DANR (1996). *Almond production manual. University of California. Division of Agriculture and Natural Resources. Pub.*, *3364*.

UC DANR (1998). *Walnut production manual. University of California. Division of Agriculture and Natural Resources. Pub.*, *3373*.

Ulaby, F. T., Li, R. Y., & Shanmugan, K. S. (1982). Crop classification using airborne radar and Landsat data. *IEEE Transactions on Geosciences and Remote Sensing*, *20*, 518−528.

USDA-NASS (1997). Usual planting and harvesting dates for U.S. field crops.http://www.nass.usda.gov (last accessed October 3, 2010).

USDA-NASS (2006). Crop progress and condition in California.http://www.nass.usda.gov (last accessed October 3, 2010).

van Deventer, A. P., Ward, A. P., Gowda, P. H., & Lyon, J. G. (1997). Using Thematic Mapper data to identify contrasting soil plains to tillage practices. *Photogrammetric Engineering and Remote Sensing*, *63*, 87−93.

Wardlow, B. D., & Egbert, S. L. (2008). Large-area crop mapping using time-series MODIS 250 m NDVI data: An assessment for the U.S. Central Great Plains. *Remote Sensing of Environment*, *112*(3), 1096−1116.

Wardlow, B. D., Egbert, S. L., & Kastens, J. H. (2007). Analysis of time-series MODIS 250 m vegetation index data for crop classification in the U.S. Central Great Plains. *Remote Sensing of Environment*, *108*(3), 290−310.

Wright, C., & Gallant, A. (2007). Improved wetland remote sensing in Yellowstone National Park using classification trees to combine TM imagery and ancillary environmental data. *Remote Sensing of Environment*, *107*(4), 582−605.

Xu, M., Watanachaturaporn, P., Varshney, P. K., & Arora, M. K. (2005). Decision tree regression for soft classification of remote sensing data. *Remote Sensing of Environment*, *97*(3), 322−336.

Yolo County Agriculture Department (2006). Agricultural Crop Report.http://www.yolocounty.org/Index.aspx?page=1523 (last accessed October 3, 2010).

Zhang, Y. (1996). A survey on evaluation methods for image segmentation. *Pattern Recognition*, *29*(8), 1335−1346.

Zheng, H., Chen, L., Han, X., Zhao, X., & Ma, Y. (2009). Classification and regression tree (CART) for analysis of soybean yield variability among fields in Northeast China: The importance of phosphorus application rates under drought conditions. *Agriculture, Ecosystems & Environment*, *132*(1–2), 98−105.